6.S974 class notes

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Contents

Games

Normal-form game. *[not covered in HW2]*

Static Bayesian game (HW2 P1). Each player has a private type θ_i drawn from a common prior. Players move at the same time. A strategy is one action per type, $s_i : \Theta_i \to A_i$. The opponent responds to the type-weighted probability of each action, $\alpha = \sum_\theta P(\theta)\, \sigma_i(a \mid \theta)$.

Signaling game (HW2 P2). The sender knows its type and sends a message m. The receiver sees m, not the type, and acts. The receiver has one information set per message. The sender has one per type.

Equilibria

Nash equilibrium. *[not covered in HW2]*

Bayesian Nash equilibrium (HW2 P1). Object: σ^*. Check, for every player i and type θ_i:

$$\operatorname{supp} \sigma_i^*(\cdot \mid \theta_i) \subseteq BR_i(\sigma_{-i}^* \mid \theta_i).$$

Pure BNE: candidates are pure profiles; a best-response cycle rules out its members. Mixed BNE: a type mixes only if indifferent, and its indifference fixes the opponent's α at a threshold. Verification: list the check for every type, both sets written out.

Weak perfect Bayesian equilibrium (HW2 P2). Object: (σ^*, μ^*), including any off-path beliefs you chose. Check 1, for every information set I:

$$\operatorname{supp} \sigma_i^*(\cdot \mid I) \subseteq BR_i\big(\mu_i^*(\cdot \mid I),\, \sigma_{-i}^*\big|_I\big).$$

Check 2, for every on-path information set: μ^* follows from the prior and σ^* by Bayes' rule. Off-path beliefs are chosen, not derived. Verification: both checks at every information set. Beliefs feed the receiver; the receiver feeds the sender; fill in that order.

Separating. Each type sends a different m. Both receiver sets on-path, beliefs 0 or 1. Check each type against the other's message.

Pooling. Both types send the same m. Belief after m is the prior. The unsent message is off-path; choose its belief so the deterring action is a best response.

Semi-separating. One type pure, one type mixes with p. Receiver indifference after m fixes p. Receiver then mixes with q; sender indifference fixes q. Check the pure type.